

1. Introduction

- Vowels are **shorter and more likely to devoice** in these contexts:

- when they are **[+high]** (Lehiste 1970, Solé and Ohala 2010)
- when they are before **C₂[-voice]** (Chen 1970)

Is the short duration of devoiced vowels the **cause (H1)** or the **result (H2)** of devoicing?

H1: V_[+high], C₂[-voice] → Duration shortening → Vowel devoicing

- Vowel devoicing is a result of gestural overlap (Jun et al. 1997)
- Contexts affect vowel devoicing, mostly via vowel shortening

H2: V_[+high], C₂[-voice] → Vowel devoicing → Duration shortening

- In Tokyo Japanese, while both C₁[-voice]V_[±high] shorten before C₂[-voice], C₁[-voice]V_[+high] before C₂[-voice] shortens much more (Tanner et al. 2019) because high vowels devoice in that context
- Contexts affect vowel shortening mostly via vowel devoicing

RQ1. Is duration of C₁[-voi]V_[+hi] shorter than what is expected from the intrinsic duration of V_[+hi] and shortening before C₂[-voi]?

RQ2. Is the likelihood of VD correlated with CV duration?

RQ3. Does devoicing mediate context effects on CV duration?

2. Methods

Tokens of interest: C₁[-voice] V_[±high] C₂[±voice]

Tokyo Japanese (TJ) (n = 116,876)

- Corpus of Spontaneous Japanese** (Maekawa et al. 2000)

- C₂[+voice] = {voiced obstruents, sonorants}
- C₂[-voice] = {voiceless obstruents}
- V devoicing annotated in the corpus
- Long V and geminate C are excluded

Seoul Korean (SK) (n = 50,349)

- Seoul Corpus** (Yun et al. 2015)

- C₂[+voice] = {voiced lenis obstruents, sonorants}
- C₂[-voice] = {voiceless fortis/aspirated obstruents}
- V devoicing, lenis voicing determined by Voice Report in Praat

Baysian mixed-effects regression models

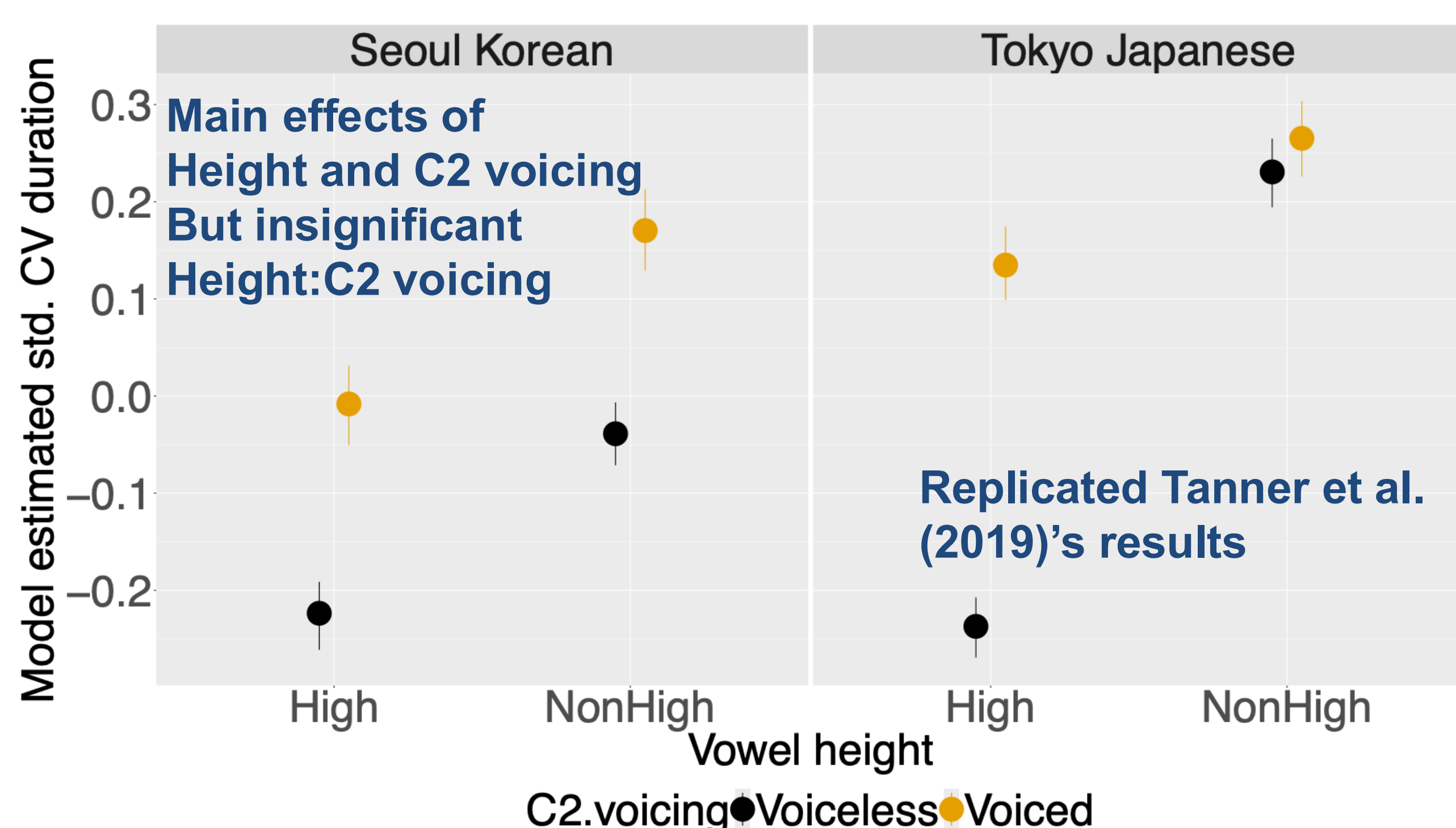
- DV: Vowel devoicing (Bernoulli with logit link), CV duration (Gaussian)
- IV: V height (binary), C2 voicing (binary)
- Covariates: Preceding C manner, Following C manner, speech rate
- by-speaker and by-word random intercepts and random slopes

Mediation analysis:

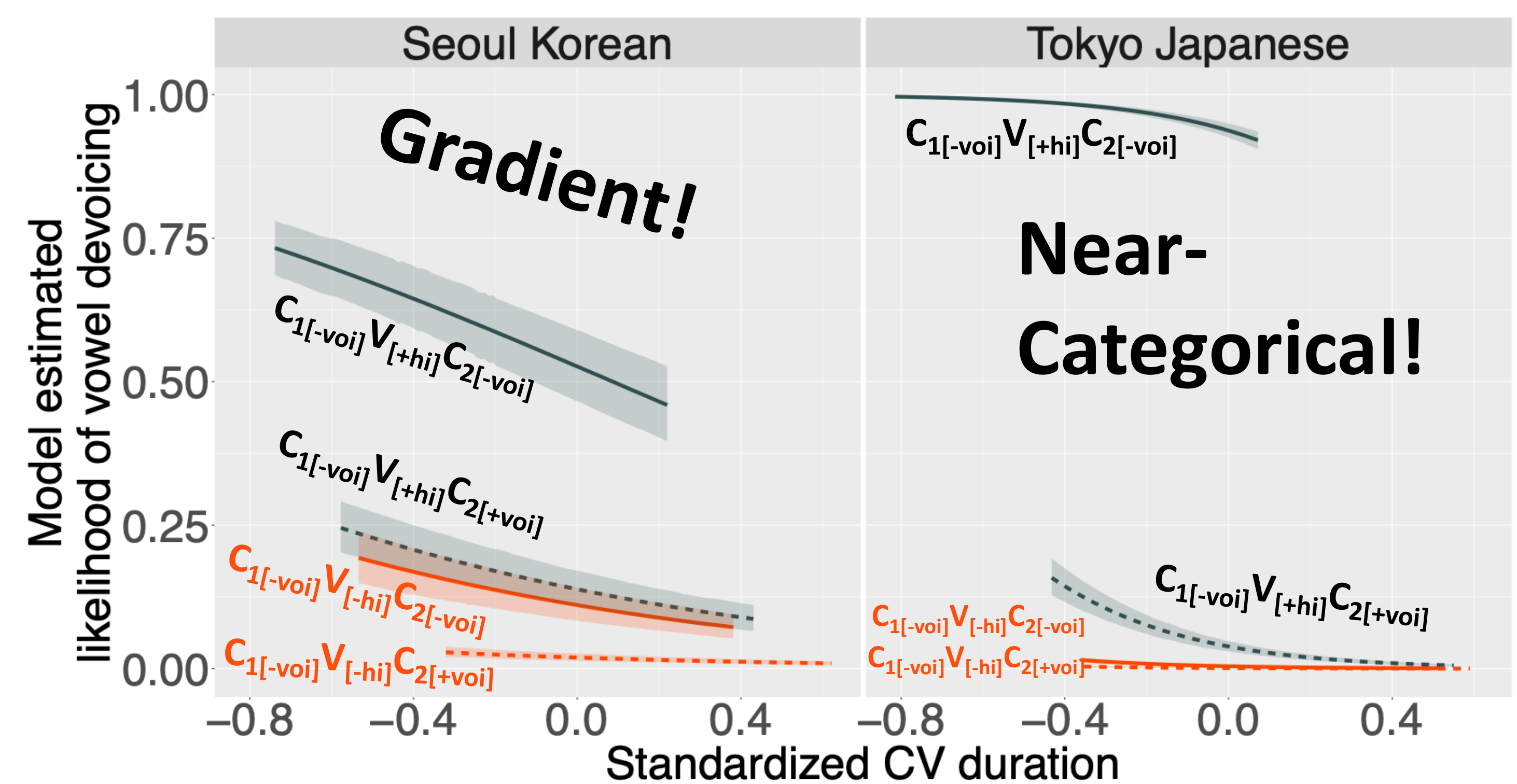
Compares regression models fit with and without a mediator variable (Cohen Priva and Gleason 2020, Katz and Pitzanti 2019, Lee 2025)

If context *causes* duration shortening mostly via vowel devoicing (H2), then the **indirect effect** on CV duration via vowel devoicing should be **similar** to the **total effect** of contexts on CV duration

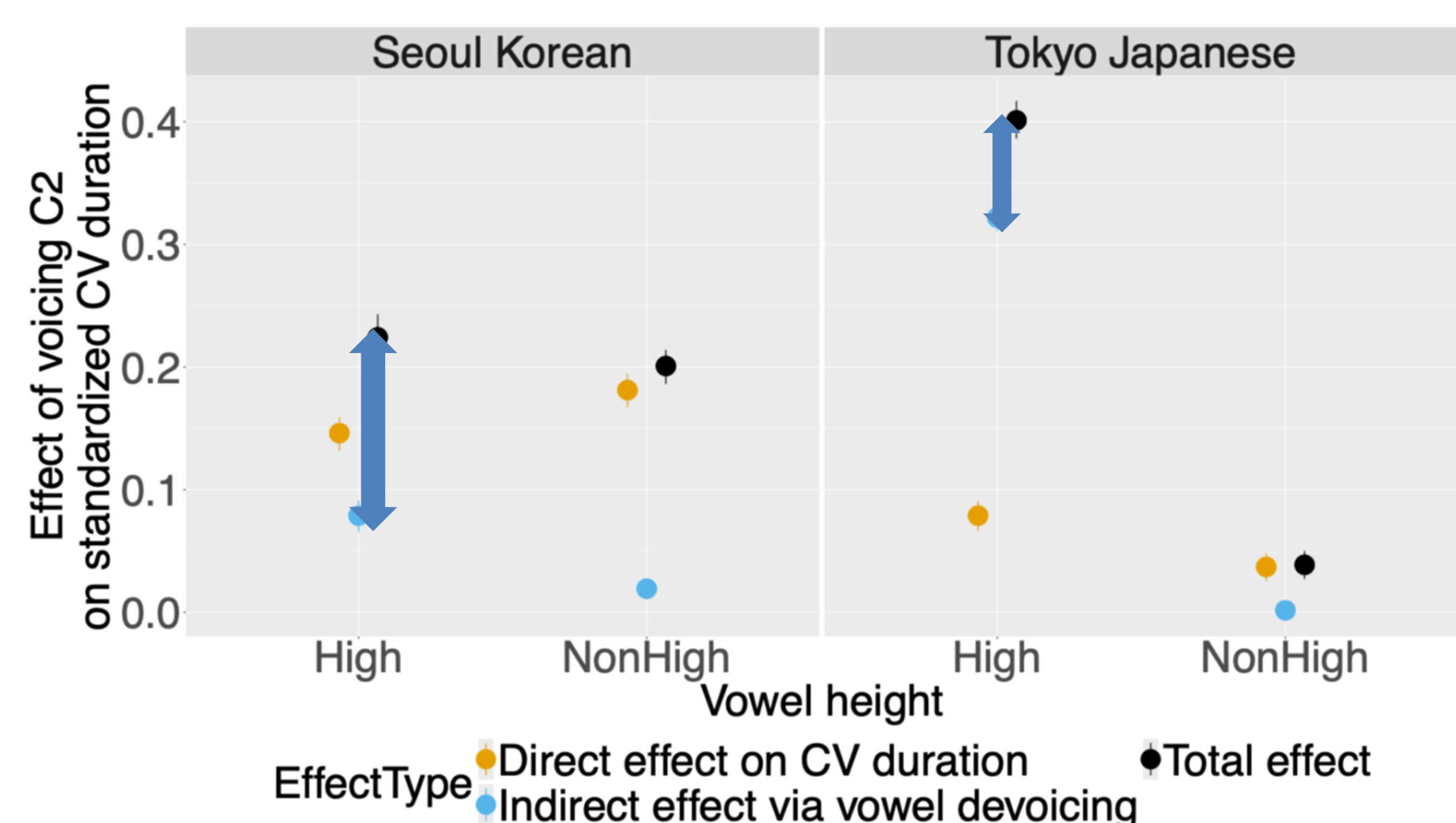
3.1. CV duration



3.2. CV duration x Devoicing



3.3. Mediation analysis



4. Discussion

TJ: Strong support for H2 (Context → VD → Duration shortening)
SK: Weak support for H1 (Context → Duration shortening → VD)

- Potential reason for the difference between SK and TJ: **Phonologization**
- Coarticulation as a precursor to a phonological process (Beddor 2009)
- At the **phonetic** level, VD occurs as a result of coarticulatory shortening (and other factors), as predicted by gestural overlap (H1)
- But once VD is **phonologized**, short duration of CV cannot be attributed to coarticulatory shortening (H2)
- Phonologization may change the causal structure of a process and its associated phonetic variables (H1 → H2)

5. Conclusion

- Variation in CV duration** can be explained by general coarticulatory shortening in SK, but **not** in TJ.
- The likelihood of VD** can be explained by coarticulatory shortening in SK to some extent, but **not** in TJ.
- The difference could be due to **phonologization** —VD is phonologized in TJ while VD is at the phonetic level in SK.